

**A
Project Report
on
"Paralyze patient health monitoring system"**

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This is to certify that the report entitled “**Paralyze patient health monitoring system**” is a bonafide work carried out by **Tanmay Patel (19EC056)** under the guidance and supervision of **Prof. Miral Desai** for the subject **Project (EC458)** of 8th Semester of Bachelor of Technology in Electronics & Communication at Faculty of Technology & Engineering (C.S.P.I.T.) – CHARUSAT, Gujarat.

To the best of my knowledge and belief, this work embodies the work of candidate himself, has duly been completed, and fulfills the requirement of the ordinance relating to the B.Tech. Degree of the University and is up to the standard in respect of content, presentation and language for being referred to the examiner.

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1. Abstract

It is well known that paralysis is the term used to describe the inability of particular body parts' muscles to operate. It can happen at any time and without any pain in the affected location. The quality of life for those who are paralyzed has improved thanks to technical and therapeutic advances. Our goal is to develop an accessible, user-friendly gadget with a fundamental health monitoring system and nursing care or home care. Our technology was created to make it possible for paralyzed people to show messages using basic motions because they frequently struggle to communicate their requirements or messages.

The Paralyzed Patient Health Monitoring System is a comprehensive healthcare solution that leverages Internet of Things (IoT) technology to monitor the health status of paralyzed patients. The system is designed to measure and record vital signs such as pulse rate and temperature, as well as body movements using an MPU6050 sensor. The data is then displayed on a 16x2 LCD display for local monitoring and is also uploaded to the Blynk Cloud through the internet for remote monitoring. The system is easy to use and provides a cost-effective solution for monitoring paralyzed patients in both hospital and home care settings.

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Abbreviations

IoT	Internet of Things
LoRaWAN	Long Range Wide Area Network

CHAPTER 1:- Introduction

1.1 Objective

The objective of the Paralyzed Patient Monitoring System project is to develop a comprehensive healthcare solution that utilizes IoT technology to monitor the wellbeing of patients who are paralyzed. The system is designed to track and record bodily motions, as well as vital indications such as temperature and heart rate, using sensors and microcontrollers. The data is then processed and analyzed in the cloud, and caregivers can remotely monitor the patient's condition through a user interface. The project aims to provide a reliable and efficient way to monitor the health of paralyzed patients, ensuring timely medical intervention if needed and improving the overall quality of care.

1.2 Motivation

To improve the quality of life for patients who are paralyzed: The monitoring system provides a way to track and monitor vital signs of patients who are paralyzed. This can help to detect any health issues early on and provide prompt medical attention, improving the quality of life for the patients [11].

To provide peace of mind to caregivers and family members: Caregivers and family members of patients who are paralyzed often worry about the patient's health and safety. The monitoring system provides a way for them to remotely monitor the patient's vital signs, giving them peace of mind.

To reduce healthcare costs: Early detection and intervention of health issues can reduce healthcare costs in the long run [10]. The monitoring system can help to detect health issues early so, early treatment potentially reducing healthcare costs associated with hospitalizations or emergency care.

1.3 Introduction

In 21st century, there are crucial requirement of various types of parameters which is sensed by different types of sensor. Sensors work as input of the data which is performing by main system. Sometimes it is required to keeping log and doing analysis of these parameters and has to take some necessary control action according to the specific behavior of this measured value. We also need that kind of system which is work smartly but complete smart system has some restrictions. So, to reduce that kind of restrictions we need hybrid smart system which is working as smart system but when human need to take control at that time it will give option to humans to operate the things over internet. All the things can happen only happens if the system is well configured and fast Internet service. The suggested system is based on IOT Blynk cloud platform. The reason to design the system is based on the Blynk cloud platform to send the data over the internet and store/monitor remotely as well as the control whenever needs. The advantage of this system is we can monitor the data and control connected respective devices from anywhere of the world through internet. It can send a notification when detect some unnecessary movements by the connected respective sensors. Different connected sensors connect different quantities and after sensing the parameters it give appropriate action to the actuators or give the control to the cloud widgets. In proposed system one dedicated task is running. The 1st task is running and printing messages as per the rotation of the gyroscope sensor. The 2nd task is to send the received data to the cloud.

1.4 Literature Review

Now days Smart City is under IOT Technology are becoming more and more popular. Main aim of the IOT is connect hardware to internet. Then, Web on Things (IOT) emerged to easily connect sensors to web. Print the data and exchange the data on web that has been produced by the device.

A health monitoring system is an electronic device that collects and processes medical data from a patient and transmits it to a healthcare professional for analysis [5]. It can include sensors, wireless communication technology, and data analytics software to monitor various vital signs such as heart rate, blood pressure, body temperature, and oxygen saturation. The system is designed to provide continuous, real-time monitoring of a patient's health and allows healthcare providers to track the progression of the patient's condition [12]. The use of health monitoring systems has become particularly important during the COVID-19 pandemic, as it allows healthcare providers to monitor patients remotely and reduce the risk of infection transmission [1].

The main motive of "Paralyzed Patient Health Monitoring System" is to develop a comprehensive healthcare solution using IoT technology that can keep track of the wellbeing of patients who are paralyzed [1]. The system uses an MPU6050 sensor to track and record bodily motions as well as vital indications including temperature and heart rate. The information is then shown on a 16x2 LCD display for local monitoring and is further transferred through the internet to the Blynk Cloud for carer monitoring from a distance. The ESP32 microcontroller serves as the system's central component, offering wireless communication to the Blynk Cloud.

This project offers less cost as compare to the other system which is using the BLE technology. The main advantage of this system is to monitor the data locally as well as the globally using internet of things while, BLE has limited range of communication. For that reason this system is much more convincing than other system.

Overall, the literature review suggests that IoT-based patient monitoring systems have gained popularity due to their ability to provide real-time monitoring and feedback, allowing for timely intervention and better healthcare outcomes.

CHAPTER 2:- IoT

2.1Introduction

The Internet of Things (IoT) refers to the network of physical devices, vehicles, home appliances, and other objects that are embedded with sensors, software, and connectivity, allowing them to collect and exchange data with each other and with other systems over the internet. The IoT is rapidly transforming the way we live and work, offering a range of benefits and opportunities for individuals, businesses, and society as a whole.

One of the key benefits of the IoT is its ability to provide real-time data and insights that can be used to improve efficiency, productivity, and decision-making [6]. For example, in the healthcare sector, IoT devices can be used to monitor patients' health and provide alerts to medical professionals if any issues arise, allowing for early intervention and better outcomes. In the manufacturing industry, IoT sensors can be used to track the performance of equipment and identify areas for optimization, reducing downtime and improving efficiency.

The IoT is also enabling new business models and revenue streams, as companies can use the data generated by IoT devices to create new products and services, and to gain deeper insights into customer behavior and preferences. For example, a smart home system could collect data on a homeowner's energy usage patterns, and use this information to offer personalized recommendations for reducing energy consumption and lowering costs.

However, the widespread adoption of the IoT also raises concerns about data privacy and security. With so much data being collected and shared, it is essential that proper security measures are in place to protect against unauthorized access and data breaches [8].

Despite these challenges, the IoT is expected to continue to grow and evolve in the coming years, as more devices become connected and new

applications are developed. It has the potential to transform virtually every aspect of our lives, from how we live and work to how we interact with the world around us. As such, it is essential that we work to address the challenges and opportunities presented by the IoT, in order to fully realize its potential and ensure that it benefits society as a whole.

2.2IoT DESIGN FLOW

IoT flow design is a process of designing the communication and data flow architecture for an IoT system. It involves creating a blueprint of how the IoT devices, sensors, gateways, and cloud services will interact with each other to achieve the desired functionality [9].

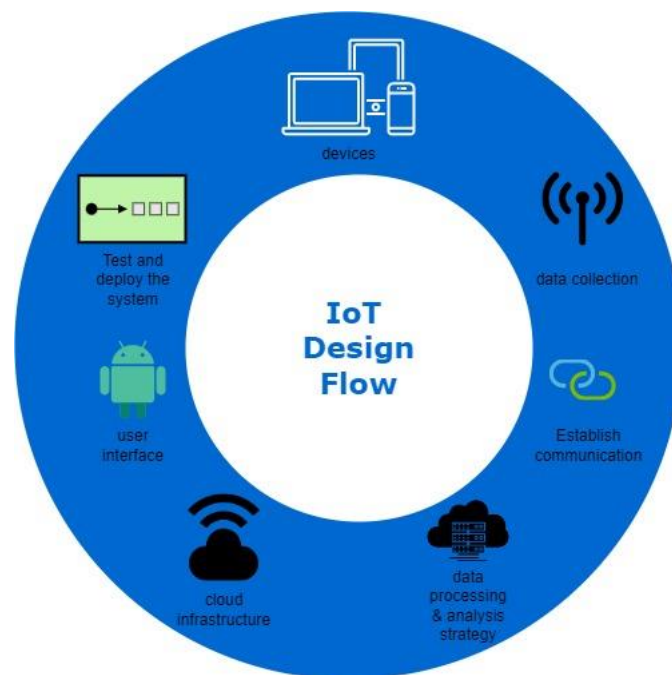


Fig. 1.1 IoT Design Flow

1. **Identify the IoT devices and sensors:** IoT devices and sensors are the fundamental building blocks of the Internet of Things (IoT) ecosystem. These devices and sensors are used to collect data from the physical world and send it to the cloud or a local server, where it can be processed, analyzed, and acted upon.

- 1) **IoT devices:** IoT devices are physical devices that can connect to the internet and communicate with other devices or systems. These devices can range from simple sensors to complex devices with embedded processors and connectivity modules. Some of the most common types of IoT devices include smart thermostats, home security systems, wearable devices, smart lighting, and smart locks.
- 2) **Sensors:** Sensors are electronic devices that can detect and measure physical parameters such as temperature, pressure, humidity, light, sound, and motion. They are often used in IoT applications to collect data from the environment and transmit it to a central system for analysis [10]. Some of the most common types of sensors used in IoT include temperature sensors, humidity sensors, pressure sensors, accelerometer sensors, and proximity sensors.

Overall, IoT devices and sensors play a critical role in the development and deployment of IoT solutions. They enable the collection of real-time data from the physical world and the generation of insights that can be used to improve efficiency, productivity, and user experience.

2. **Define the data collection strategy:** Data collection strategy is a critical aspect of any IoT system. It involves designing a plan for capturing, processing, and analyzing data generated by IoT devices and sensors. Determine how the data from the devices will be collected [3]. This can be achieved through a variety of communication protocols such as Wi-Fi, Bluetooth, Zigbee, or cellular networks.

- 1) **Define the objectives:** The first step in developing a data collection strategy is to clearly define the objectives of the IoT system. This will help determine what data needs to be collected and how it will be used.
- 2) **Identify the relevant data:** The next step is to identify the data that is relevant to the objectives of the IoT system. This includes selecting the sensors and devices that will capture the data and determining the

frequency and granularity of data collection.

Overall, a successful data collection strategy in IoT involves a combination of technical expertise, domain knowledge, and effective communication and collaboration among stakeholders. It is important to continuously evaluate and refine the strategy as new data is collected and new insights are gained.

3. **Establish communication between devices:** Establishing a connection between IoT devices is a crucial aspect of IoT systems. There are various ways to establish connections between IoT devices, depending on the network architecture, device capabilities, and communication requirements [4]. This can be achieved through wired or wireless communication channels.

Some of the common methods for establishing connections between IoT devices are:

- 1) **Wi-Fi:** Wi-Fi is one of the most popular and commonly used connectivity options for IoT devices. It provides high-speed connectivity, making it ideal for devices that require real-time data transfer. Wi-Fi is also easy to set up and does not require any additional infrastructure.
- 2) **Bluetooth:** Bluetooth is a low-power wireless communication protocol that is commonly used for short-range communication between devices. It is ideal for devices that do not require high-speed connectivity and consume low power.
- 3) **Zigbee:** Zigbee is a low-power wireless communication protocol designed specifically for IoT devices. It provides reliable and secure communication over short distances and is ideal for applications that require low data rates.
- 4) **Cellular:** Cellular networks provide wide-area connectivity for IoT devices, making them ideal for applications that require devices to be

deployed in remote locations. Cellular connectivity is also useful for devices that require high-speed connectivity.

- 5) **LoRaWAN:** LoRaWAN is a long-range wireless communication protocol that is designed specifically for IoT devices. It provides low-power, long-range connectivity and is ideal for devices that need to be deployed over large areas.

In addition to these connectivity options, other factors such as security, scalability, and interoperability also need to be considered when establishing connections between IoT devices.

4. **Define the data processing and analysis strategy:** Data processing and analysis are critical components of IoT systems that enable organizations to make data-driven decisions and gain valuable insights into their operations. The data collected from IoT devices can be analyzed to improve efficiencies, reduce costs, and enhance customer experiences [3]. Determine how the collected data will be processed and analyzed. This can be done using cloud services, edge computing, or other data processing platforms.

- 1) **Data pre-processing:** Raw data collected from IoT devices often contains noise, missing values, and outliers. Therefore, pre-processing of data is required to clean and transform it into a suitable format for analysis.
- 2) **Data aggregation:** IoT devices generate a massive amount of data, and it's essential to aggregate and summarize the data to reduce its size and make it more manageable for analysis.
- 3) **Data storage:** The collected data needs to be stored in a secure and scalable data storage system, such as a cloud-based storage service, to ensure its availability and accessibility for analysis.
- 4) **Data analysis:** Data analysis involves applying various statistical and machine learning techniques to extract insights and patterns from the data.

This can include predictive modeling, clustering, regression analysis, and anomaly detection[3].

- 5) **Real-time data analysis:** In some IoT applications, real-time data analysis is critical to identify and respond to events as they occur. Real-time data analysis requires specialized technologies and algorithms that can process data quickly and accurately.
- 6) **Data visualization:** Data visualization is an essential aspect of IoT data analysis as it enables users to interpret and communicate complex data insights effectively.
- 7) **Data security:** Data security is a critical consideration in IoT systems, and appropriate security measures must be implemented to ensure the confidentiality, integrity, and availability of the data.

Overall, a comprehensive data processing and analysis strategy is essential to extract meaningful insights from IoT data and make informed decisions that drive business value.

5. **Create the cloud infrastructure:** Creating a cloud infrastructure is an important step in developing an IoT system, as it allows for the storage, processing, and analysis of data generated by connected devices. There are several cloud infrastructure options available for IoT systems, including public cloud platforms like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform, as well as private cloud solutions.
6. **Develop the user interface:** Developing a user interface for an IoT system is important as it provides a means for users to interact with and control the system [1]. The user interface can be developed as a web or mobile application, depending on the target platform and user requirements.

Here are some key points to consider when developing a user interface for an IoT system:

- 1) **User Experience (UX) Design:** It is important to consider the user experience when designing the user interface. The interface should be easy to use and navigate, and should provide users with relevant information and actions.
- 2) **Real-time Data Display:** The user interface should be designed to display real-time data collected by the IoT devices. This could include visualizations such as graphs, charts, and tables, which can help users understand the data and make informed decisions.
- 3) **Alerts and Notifications:** The user interface should also include alerts and notifications to keep users informed of important events or changes in the system. For example, an alert could be triggered if a sensor detects a fault or a critical parameter crosses a threshold.
- 4) **Device Control:** Depending on the system, users may need to control IoT devices from the user interface. The interface should provide controls for devices such as switches, sliders, and buttons, which can be used to turn devices on or off, adjust settings, or trigger actions.
- 5) **Authentication and Security:** The user interface should be designed with authentication and security in mind, to ensure that only authorized users can access the system and make changes to device settings or data.

Overall, developing a user interface for an IoT system requires careful consideration of user requirements, real-time data display, device control, and security measures. A well-designed user interface can make the system easier to use and more effective in achieving its intended goals.

7. **Test and deploy the system:** Testing and deployment are crucial steps in IoT development to ensure that the system works correctly in the real world.

Testing involves verifying that the IoT system works correctly under different scenarios and conditions, such as connectivity issues, sensor

failures, and unexpected data. Developers can perform functional testing, performance testing, and security testing to ensure that the system meets the desired specifications and requirements.

After testing, the system can be deployed in the real world. The deployment process involves installing the hardware and software components, configuring the system, and testing it in the real environment. Developers can monitor the system's performance and troubleshoot any issues that arise during deployment.

To ensure the smooth deployment of the system, developers should follow best practices, such as documenting the deployment process, providing user training, and creating backup and recovery plans. It is also essential to have a plan for maintaining and updating the system to ensure its long-term functionality and reliability.

CHAPTER 3:- Blynk Cloud

3.1 Introduction

Blynk is a cloud-based platform for building and managing Internet of Things (IoT) applications. It allows users to develop custom mobile applications for controlling and monitoring devices connected to the Internet [9]. With Blynk, developers can easily create user interfaces for controlling hardware such as sensors, motors, and other devices. The platform also provides a range of features for data logging, notifications, and remote access.

Blynk is designed to be easy to use, even for those with little or no programming experience [8]. It offers a drag-and-drop interface for building mobile applications and supports a wide range of hardware platforms, including Arduino, Raspberry Pi, ESP8266, and others. The Blynk platform provides a range of libraries and tools to help developers get started quickly [6].

One of the key advantages of Blynk is its cloud-based infrastructure. The platform handles all the complex backend tasks, such as data storage, security, and scaling, allowing developers to focus on building their applications. The Blynk cloud also provides a range of services, such as push notifications, SMS messaging, and email alerts, making it easy to keep track of device activity and respond to events in real-time.

Overall, Blynk is a powerful and flexible platform for building IoT applications. Its user-friendly interface, extensive library of tools and resources, and cloud-based infrastructure make it an attractive option for both novice and experienced developers.

3.2 Hardware

Blynk supports a wide range of hardware platforms including Arduino, Raspberry Pi, ESP8266, ESP32, Particle, and many more [6]. The platform offers libraries and examples for each of these devices to make it easy for developers to

get started with their projects.

In addition to these hardware platforms, Blynk also offers support for a range of sensors and actuators, such as temperature sensors, humidity sensors, motion sensors, LED displays, and more. This makes it easy for developers to quickly create IoT projects that can collect data from the physical world and interact with it in real-time.

Blynk also offers integrations with popular IoT platforms such as IFTTT, Zapier, and Microsoft Azure, making it easy to connect Blynk projects with other services and platforms. This flexibility and ease of use have made Blynk a popular choice for hobbyists and professional developers alike.

3.3 Advantages

Blynk is a powerful and user-friendly platform for IoT development that provides a range of benefits and advantages to its users.

One of the major advantages of Blynk is its ease of use. Blynk offers a drag-and-drop interface that enables users to quickly create custom IoT applications without needing to write any code. Additionally, Blynk offers a wide range of pre-built widgets and templates, which further simplifies the development process [8].

Another key advantage of Blynk is its flexibility. Blynk supports a wide range of hardware platforms, including popular boards such as Arduino, Raspberry Pi, and ESP8266. Blynk also supports a variety of communication protocols, including Wi-Fi, Bluetooth, and Ethernet, making it easy to integrate with a range of devices and systems.

Blynk also offers a robust cloud infrastructure that provides secure data storage and access to real-time analytics [2]. This allows developers to easily monitor and analyze data from connected devices, which can be used to improve application performance and identify potential issues.

Overall, Blynk is an excellent platform for IoT development that offers a

range of benefits and advantages, including ease of use, flexibility, and robust cloud infrastructure.

3.4 Disadvantages

While Blynk is a powerful and flexible platform for IoT development, it does have some limitations to keep in mind. One of the main limitations is that Blynk relies on an internet connection to function properly. If the internet connection is slow or unreliable, it can affect the performance of the Blynk app and cause delays or disconnects [11]. Additionally, Blynk is primarily designed for prototyping and small-scale projects, so it may not be the best solution for large-scale commercial projects or applications that require high levels of security.

Another limitation of Blynk is that it requires programming knowledge to set up and use effectively. While there are many resources available to help users learn how to program with Blynk, it can be a steep learning curve for beginners without any programming experience.

Finally, Blynk has some limitations in terms of its functionality and compatibility with different hardware platforms. While it supports a wide range of hardware devices and sensors, some may not be fully compatible with the Blynk platform, or may require additional configuration and customization to work properly [9].

Overall, while Blynk is a powerful and flexible IoT platform, it is important to keep these limitations in mind when evaluating it for your project or application.

CHAPTER 4:- Arduino ide

Arduino IDE (Integrated Development Environment) is an open-source platform used to write and upload code to the Arduino board. It is a software tool that is used to program the Arduino microcontroller board. Arduino IDE is compatible with various operating systems, such as Windows, Linux, and Mac OS X. It is free to download and use [7].

The Arduino IDE provides a user-friendly interface that simplifies the process of writing, compiling, and uploading code to the Arduino board. The code is written in C/C++ programming language with a simplified structure that makes it easy for beginners to understand. The IDE also includes a library of pre-written code that can be easily integrated into the user's project.

The Arduino IDE includes a text editor that allows the user to write and edit the code. The editor has features like syntax highlighting, auto-complete, and error checking that help the user write clean and error-free code.

The IDE also includes a serial monitor that allows the user to communicate with the Arduino board. It displays the output of the board and allows the user to send commands to the board. This feature is useful for debugging and testing the code.

Arduino IDE supports various Arduino boards, including Arduino Uno, Mega, Nano, and Due. The IDE also supports other boards based on the Atmel AVR and ARM processors. It provides an easy-to-use platform for programming and developing projects with these boards.

In conclusion, Arduino IDE is an essential tool for programming and developing projects with Arduino boards. It provides a user-friendly interface that simplifies the process of writing, compiling, and uploading code to the board. Its compatibility with various operating systems and support for multiple Arduino boards make it a popular choice among hobbyists and professionals alike.

5 Hardware Level Block Diagram

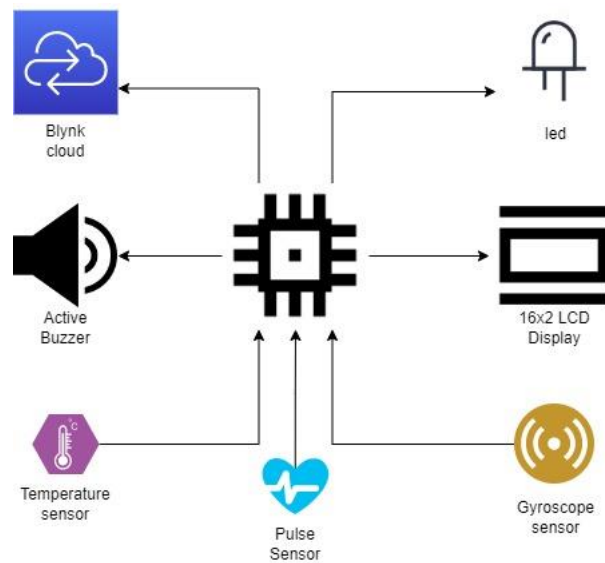


Fig. 5.1 Block Diagram

1) ESP32 Board



The ESP32 board is shown in figure. The ESP32 board is dual core processor which means it has two processor [11]. It is working on 240MHz clock frequency. The board consists of inbuilt ADC and USB connection port.

Fig. 5.2 ESP32

The technical specification of ESP32 is as follows:

- I. Microcontroller : Tensilica 32-bit Single-/Dual-core CPU Xtensa LX6
- II. Operating Voltage : 3.3V
- III. Input Voltage : 7-12V
- IV. Digital I/O Pins (DIO) : 25

- V. Analog Input Pins (ADC) : 6
- VI. Analog Outputs Pins (DAC) : 2
- VII. UARTs : 3
- VIII. SPIs : 2
- IX. I2Cs: 3
- X. Flash Memory : 4 MB
- XI. SRAM : 520 KB
- XII. Clock Speed : 240 Mhz
- XIII. Wi-Fi : IEEE 802.11 b/g/n/e/i:
- XIV. Dimensions : 51.5x29x5mm

2) 16x2 LCD Display

The 16x2 LCD display module is a commonly used alphanumeric display module in electronic projects. It consists of 16 columns and 2 rows of characters, which means it can display up to 32 characters at a time. Each



character is composed of a 5x8 dot matrix and can display a wide range of characters, including letters, numbers, symbols, and special characters.

Fig. 5.3 LCD Display

The module is relatively simple to use, and typically includes a built-in controller chip such as the HD44780 that interfaces with a microcontroller or other device to receive and display data [1]. The module can be powered by a 5V DC supply, and typically includes a backlight for improved visibility in

low-light environments.

The 16x2 LCD display module can be used in a wide range of applications, including simple display interfaces for microcontroller-based projects, digital clocks, thermometers, and more.

3) Pulse sensor

A pulse sensor is a type of sensor used to detect the heartbeat rate of an individual. It is a non-invasive sensor that can be placed on various parts of



the body such as the fingertip, earlobe, or wrist. The sensor works by using a photoplethysmography (PPG) technique, which detects changes in blood volume as it flows through the blood vessels. The PPG technique works by shining a light source onto the skin and detecting the reflected light with a photodetector.

Fig. 5.4 Pulse sensor

Pulse sensors are commonly used in healthcare and fitness applications to monitor heart rate during exercise or other physical activities. They are also used in medical devices such as pulse oximeters, which measure both heart rate and blood oxygen levels [2].

Pulse sensors can be integrated with microcontrollers or other electronic devices to provide real-time heart rate data. They are available in various form factors such as wearable devices, clip-on sensors, and finger-mounted sensors. Some pulse sensors also come with built-in algorithms to filter out noise and provide accurate heart rate readings.

4) Temperature sensor

The LM35 is a precision analog temperature sensor that can measure temperature from -55°C to $+150^{\circ}\text{C}$. It is commonly used in various applications such as temperature control systems, industrial process control, and thermal protection circuits. The LM35 sensor outputs a voltage that is



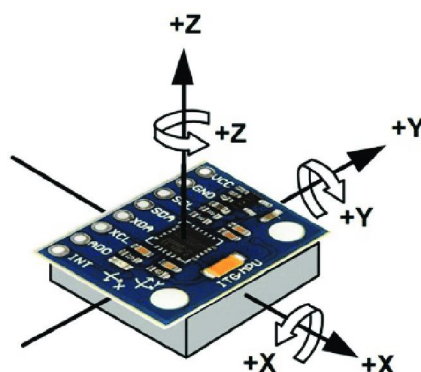
linearly proportional to the temperature being measured. Specifically, it outputs 10 mV per degree Celsius change in temperature. This linear output voltage makes it easier to interface with microcontrollers or other analog circuits for temperature measurement.

Fig. 5.5 Temperature sensor

The LM35 sensor has a low self-heating capability and a low impedance output, which ensures accuracy and stability over a wide range of temperatures. It also has a low output noise that makes it suitable for temperature measurement in high-precision applications.

5) Gyroscope sensor

A gyroscope sensor, also known as a gyro sensor, is a type of motion



sensor that measures angular velocity or rotational motion. It consists of a spinning wheel or disc that rotates at a high speed, typically in the range of tens of thousands of revolutions per minute (RPM), and is mounted on a set of gimbals that allow it to rotate freely in any direction.

Fig. 5.6 Gyroscope/Accelerometer sensor

As the gyroscope spins, it resists any change in the orientation of its axis of rotation, due to the principle of conservation of angular momentum [1]. This means that if the gyroscope is rotated around one axis, it will try to maintain its orientation with respect to that axis, even if the rest of the system is moving.

Gyro sensors are used in a wide range of applications, including navigation systems, robotics, and video game controllers. In the context of IoT, gyro sensors can be used to measure the orientation of objects, such as drones or autonomous vehicles, and to detect changes in their position or movement.

6) Active buzzer

An active buzzer module is an electronic device that produces a sound or tone when an electrical signal is applied. It is a type of piezoelectric buzzer that generates sound by using a ceramic element that vibrates when an alternating current (AC) is applied to it. The active buzzer module has a built-in oscillator circuit that generates the AC signal needed to drive the ceramic element, eliminating the need for an external oscillator circuit [6].



Fig. 5.7 Active Buzzer Module

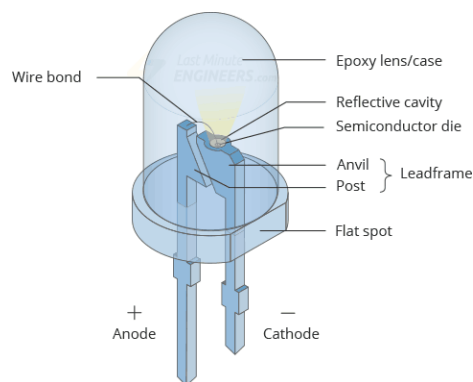
The module typically includes a small PCB with the buzzer element and a driver circuit, which can be powered by a DC voltage source such as a battery or a power supply. The module is designed to be easily integrated into electronic circuits and projects, and can be controlled by a microcontroller or other digital logic circuit. The tone or frequency of the sound produced by the buzzer can be controlled by varying the frequency of the AC signal applied to the module.

Active buzzer modules are commonly used in various applications, such

as alarms, timers, and notification systems in electronic devices and appliances.

7) Led

LED stands for Light Emitting Diode, which is a semiconductor device that emits light when an electric current is passed through it. LEDs are widely used in electronic devices, lighting, and display applications due to their low power consumption, long lifespan, and high efficiency.



An LED typically consists of a tiny chip of semiconducting material, typically a compound of gallium, arsenic, and phosphorus, which is encased in a plastic or epoxy resin. When a voltage is applied across the chip, electrons flow into the material and release energy in the form of light.

Fig. 5.8 LED

6 Software Flow Diagram

The software environment is created using Arduino C language and blynk server. The detail software flow diagram for suggested system is shown in below fig.

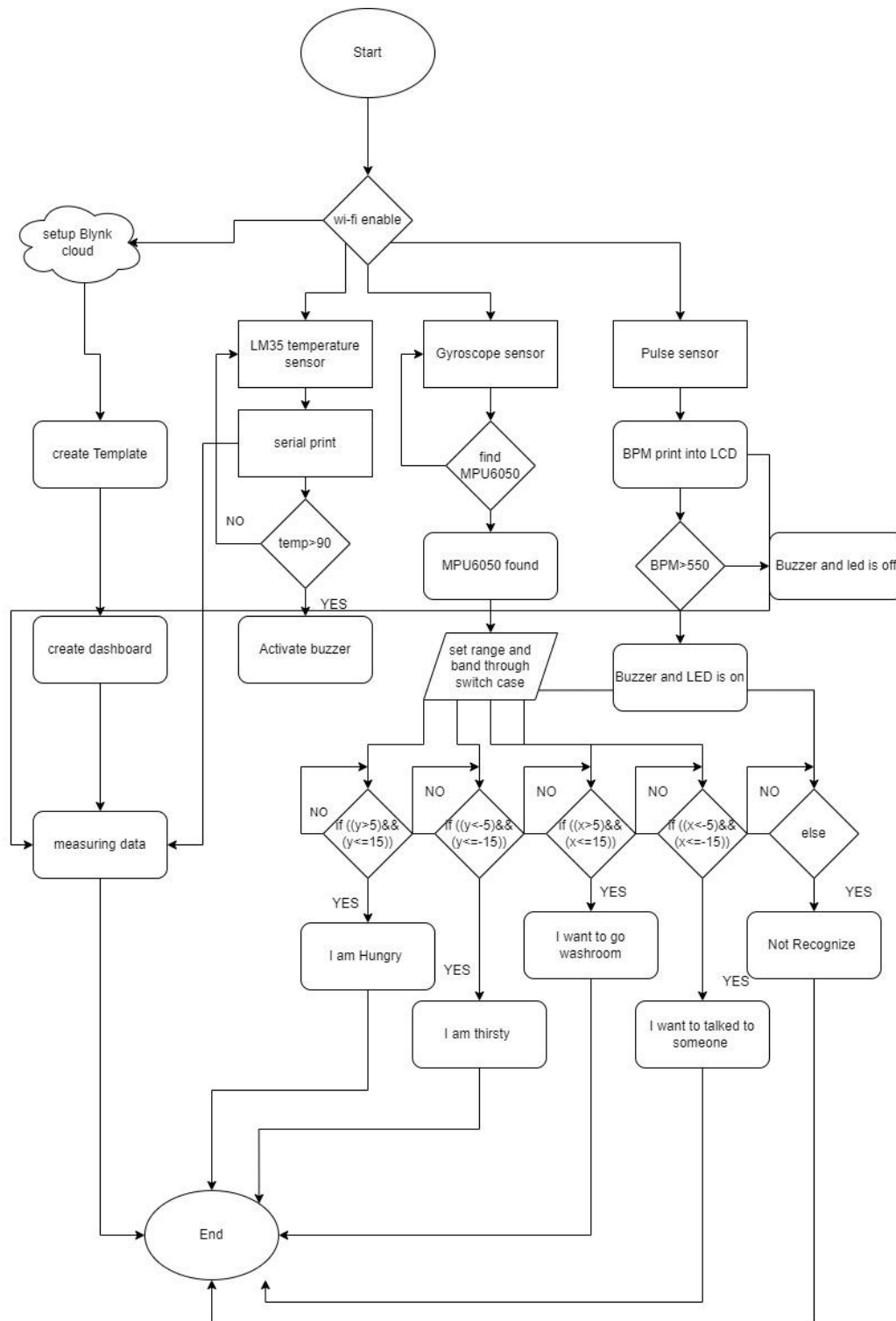


Fig. 6.1 Program Flow Chart

7 Test Results & Discussion

Experimental setup of the suggested system is in below fig. It consist of ESP32 for connect the system with Blynk server.

In this system ESP32 is a heart of the system which is used to program the system and also provide the benefit of the wi-fi, Bluetooth, BLE, etc. the ESP32 is connect with the 16x2 LCD display, gyroscope sensor, temperature sensor, LED and buzzer.

After interfacing gyroscope sensor to ESP32 and program that this gyroscope sensor can automatically find and set their accelerometer and gyroscope sensor range as well as their bandwidth. It gives the angular velocity and rotations so; set the different condition for different communication messages and watch them on 16x2 LCD display.

The active buzzer module is connected to notify local area person about patient's health.

The temperature sensor also connects with ESP32 which gives the patient body temperature to measure in real time and also active the buzzer with 2000 Hz frequency which indicate that the patients feels fever.

The pulse sensor read the patient's BPM and show on real time on blynk cloud and also the BPM is higher than the threshold then the led will glow and buzzer is active with 1000 Hz frequency.

This proposed system is used IoT technology to monitoring and communicating with other person. It can also upgraded by wireless send message and alarms to the other hardware so that we can increase the local area monitoring.

The Paralysis Patient Health Monitoring System using IoT offers several advantages over traditional monitoring systems. It provides real-time data collection and monitoring, which allows for timely intervention in case of emergencies. The system is also cost-effective and can be easily set up and

configured, making it accessible to a wider range of patients.

Moreover, the system can provide a convenient means of tracking the patient's progress and improving their quality of life. By continuously monitoring their vital signs, the system can help doctors and caregivers to make informed decisions about the patient's treatment and care.

1) Need to go washroom

The microcontroller has now displayed there later message on the LCD panel. As a consequence, if there is movement in the patient with certain angle it means they want to go washroom and if the pulse rate is high then the led will glow. As per image will be shown.

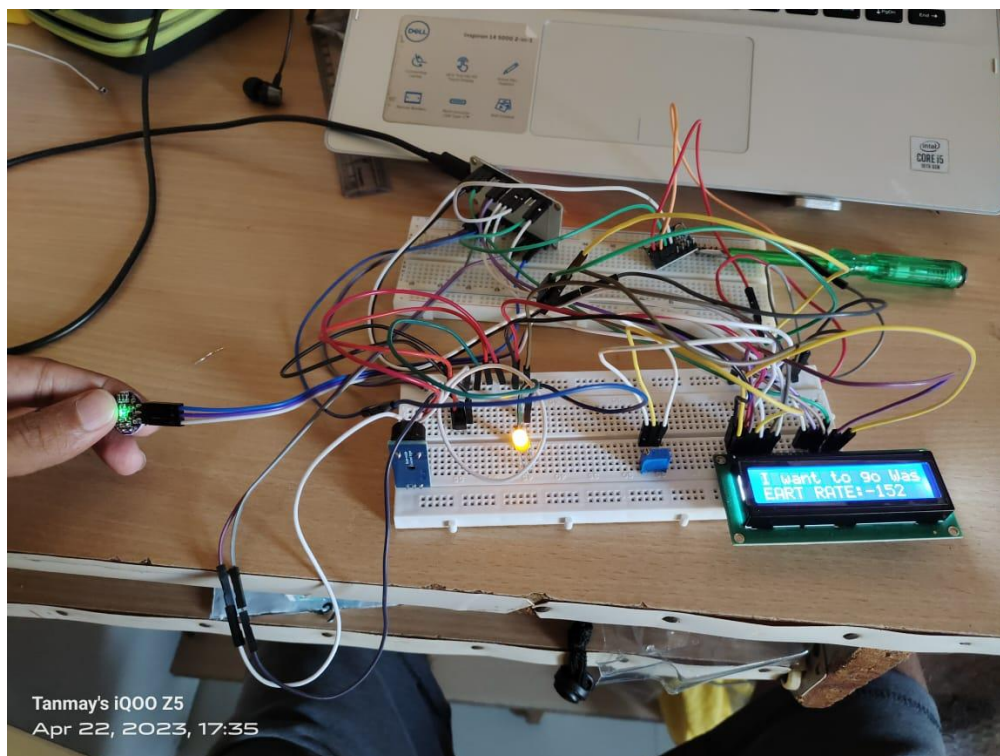


Fig. 7.1 Output 1

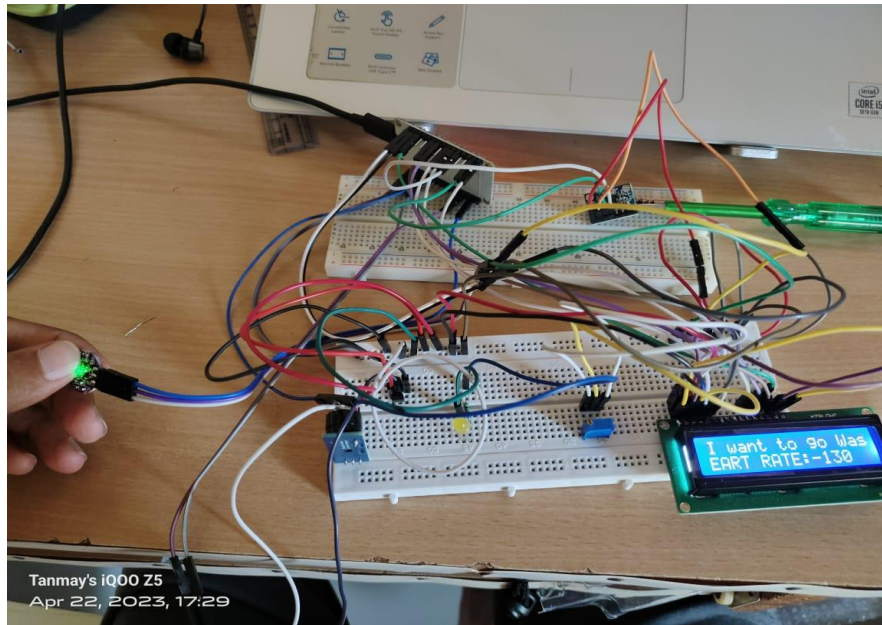


Fig. 7.2 Output 2

2) Want to talk with someone

The performance of the system is based on the angle of accelerometer. It is programmed as per the need of this patient. If it rotates in certain angle then it will show the message as “I want to talk with someone”.

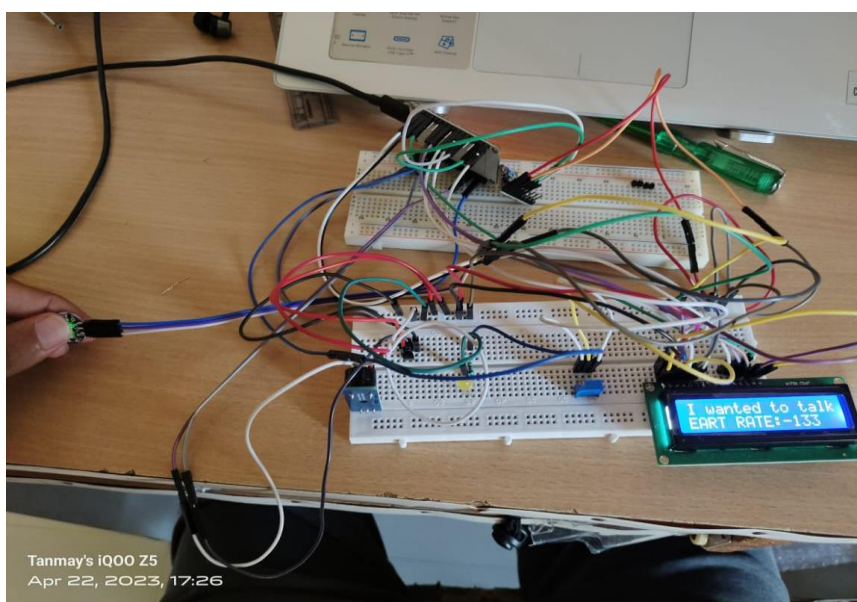


Fig. 7.3 Output 3

3) I am Thirsty

The outputs that are displayed will help patients take care of their needs. If the angle is altered in response to their demand, such as a need for water, it will alert the people. Lcd display is not just showing the message it will also shows patients pulse rate.

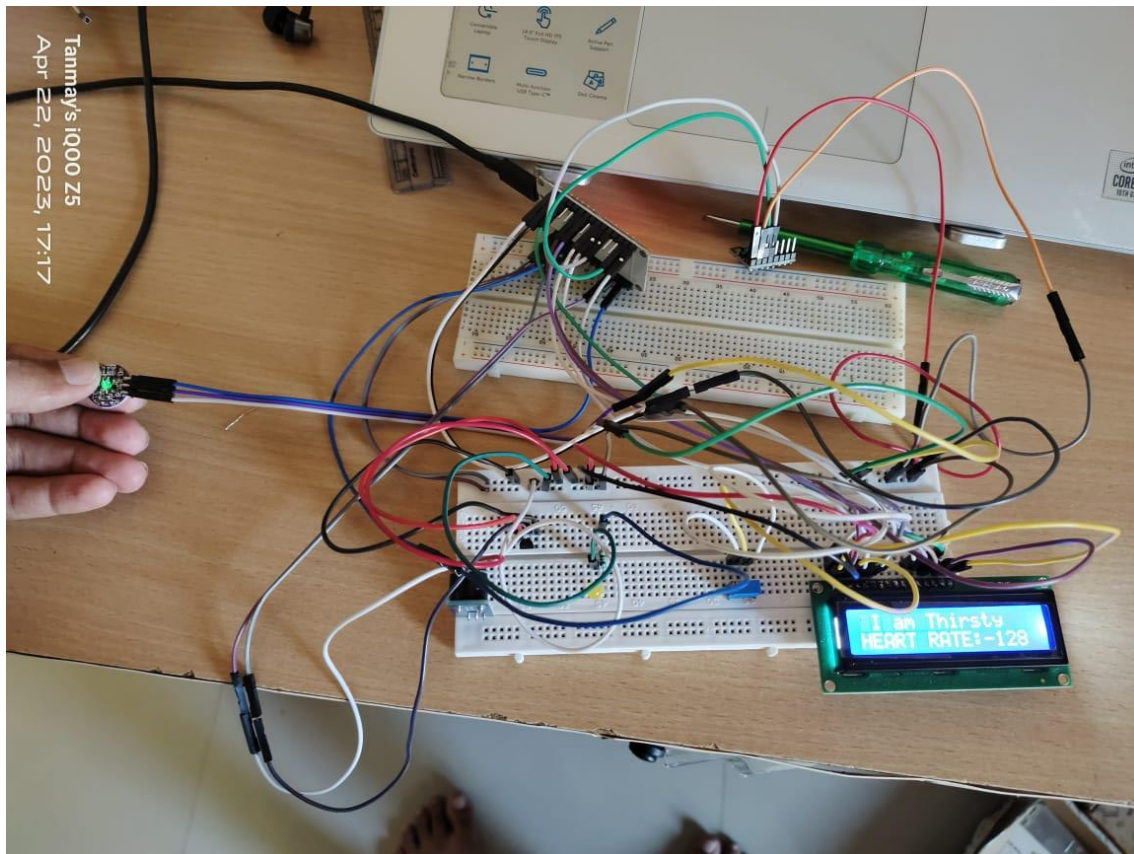


Fig. 7.4 Output 4

4) I am Hungry

The patient's condition will be closely watched at all times. If they need food, they can tilt their hand in a specific direction, and the LCD display will indicate this.

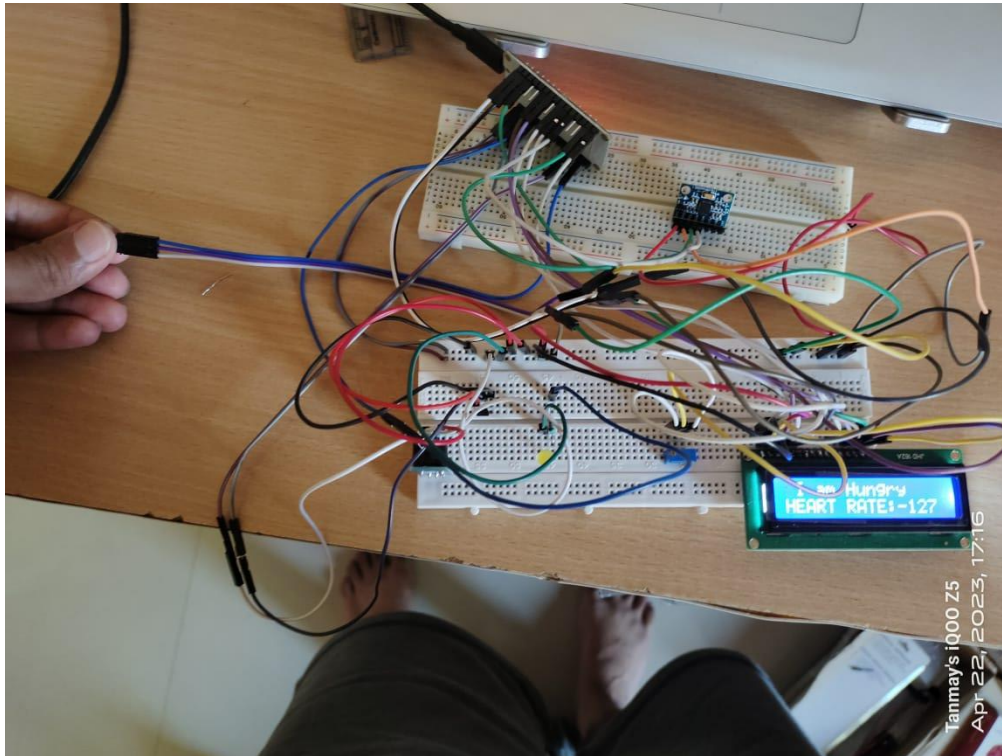


Fig. 7.5 Output 5

8 Conclusion

The Paralysis Patient Monitoring System using IoT is an innovative and effective solution for monitoring and tracking the health of paralysis patients. The system provides real-time monitoring of key vital signs such as temperature, heart rate, and blood oxygen level, along with movement data through the use of an accelerometer and gyroscope. The data is collected and processed through a microcontroller and transmitted to a cloud server via Wi-Fi, where it can be accessed remotely through a web-based interface.

The implementation of the system showed promising results, with accurate and reliable data collection and transmission. The system provides an efficient and cost-effective solution for monitoring the health of paralysis patients, allowing for timely intervention in case of emergencies [5]. It also provides a convenient means of tracking the patient's progress and improving their quality of life.

Overall, the Paralysis Patient Monitoring System using IoT has the potential to revolutionize the healthcare industry by providing a reliable and effective means of remote patient monitoring. Further development and refinement of the system can lead to improved patient outcomes and better healthcare management.

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